

Milestone One: Global Climate Impact on US Agriculture

DX799 01 Data Science Capstone

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Problem Statement:

Climate change is a serious and complex problem that has been plaguing the world since the inception of the industrial age. Humanity has exploited and produced emissions from carbon sources since early times, but has massively been accelerating our use of these resources since the 1700s (US Department of Energy, 2026). Coal, Oil, and Gas make up the majority of power generation, even in developed countries (International Energy Authority, 2025). While the process of moving to renewable energy has begun, the transition has been slow and not necessarily an important issue to most world leaders. The geopolitics of propping up the fossil fuel industry have been stifling innovation in that area, and its impacts have the potential to be catastrophic.

The ramifications of fossil fuel usage is a slow, but noticeable, change in global climate. Temperatures are on the rise, catastrophic weather events are increasing in most parts of the world, and industries have been impacted negatively as a result. Agriculture in specific is extremely dependent on climate for crop yield and drought mitigation. While we are unsure to what extent the climate is repairable, we can explore the impacts of CO₂ emissions on agriculture.

Modeling Techniques: Breadth

The two datasets used for this analysis are the CO₂ emissions per capita dataset (further referred to as the CO₂ emissions dataset) and the Climate Change impact on Agriculture dataset (further referred to as the agriculture dataset). The CO₂ emissions dataset is published by Ritchie et al. in 2024 and observes the trends in how much CO₂ is produced per capita grouped by country and region. This dataset determines which countries are producing high levels of CO₂, as well as the trends from recent years. The agriculture dataset was posted to Kaggle in

2024. This dataset combines both agricultural data along with climate metrics such as average temperature, precipitation, and extreme weather events.

The initial analysis of both datasets was using linear regression with a focus on polynomial regression. The datasets were fit to a polynomial model (degrees 1,2, and 3). R^2 value and RMSE were reported. For the CO2 emissions dataset, the target variable was Temperature Change from CO2, which was found to be highly correlated with the other variables from exploratory data analysis in a correlation matrix. The CO2 model fits best to a cubic polynomial regression, reporting a 0.0202 RMSE and 0.9189 R^2 as reported in FIG 1. For the agriculture dataset, the linear model fit the best, reporting a 0.6641 RMSE and a 0.5276 R^2 .

Variance Inflation Factor (VIF) was calculated for both datasets. This is a metric that tests for multicollinearity among variables present within each dataset, and can provide insight into whether a particular variable is too correlated with many other variables (Penn State, 2018). The VIF cutoff was determined to be < 5 . For the CO2 emissions dataset, VIF showed that none of the variables were above five, so they are not multicollinear. For the agriculture dataset, it was found that variables such as year, soil health index, irrigation access, and total precipitation had a VIF score of >5 , as shown in FIG. 2. These variables are thus collinear and are therefore unsuitable for future analysis.

Lasso, ridge, and elastic net models were used on both datasets to determine the effects of loss functions on the linear regression. In the CO2 emissions dataset, the lasso and elastic net models slightly underperformed the ridge model which reported a higher R^2 and lower RMSE. The agricultural dataset also saw ridge perform the best of the three models, as seen in FIG. 3. The very close values among lasso, ridge, and elastic net suggest that the data is well controlled

for multicollinearity and does not require additional feature selection. This is a positive sign for further analysis of the datasets.

Forward and backwards selection were performed on both datasets. For the agricultural dataset forward selection, the best R^2 and RMSE were reported when Fertilizer Use (KG per HA) was added to the dataset. The optimal backwards selection feature was removal of pesticide use (KG per HA). For the CO2 emissions forward selection, the best forward feature addition was `co2_per_capita`. The best backwards feature subtraction was `co2_per_gdp`.

Principal Component Regression (PCR) and Partial Least Squares Regression (PLSR) were performed on both datasets. Datasets were scaled with standard scalers to ensure that PCR and PLSR can be calculated accurately. R^2 and RMSE were both reported. For the agricultural dataset, PLSR outperformed PCR in both R^2 (0.5271) and RMSE (0.6645) as shown in FIG. 4. This implies that the relationships between the variables and the target are complex. The CO2 Emissions dataset had extremely close averages, indicating a strong linear relationship between the variables and the target. This is consistent with previous analysis, which shows a strong correlation between variables and the target Temperature Change from CO2 Emissions.

Logistic regression was performed on both datasets. A standard scaler was utilized to standardize the data before regression. A Receiver Operating Characteristic (ROC) curve was used to evaluate the performance of the logistic regression. The closer the AOC is to 1, the stronger the fit of the logistic regression model. For both datasets, the ROC curve performed well: Agriculture AOC: 0.85, CO2 Emissions: 0.92 (shown in FIG.5). These both show that logistic regression is not reporting many false positives or false negatives. This further reinforces that the multicollinearity has been largely eliminated from this dataset.

Support Vector Regression was performed on both datasets. A standard scaler was utilized to standardize the data before regression. The agricultural dataset has central predictive power but does not do well on both of the extremes, as seen in FIG. 6. Hyperparameter optimization was performed on C, gamma, and epsilon, but did not significantly improve the outcome. The support vector for the CO2 emissions dataset did not draw any significant conclusions. The R2 is negative, which is indicative of an overfit. This indicates that support vectors are not a good metric to evaluate this dataset with.

Modeling Techniques: Depth

The Random Forest model was chosen for in depth analysis. The random forest model is based on decision trees, and benefits from a reduced risk of overfitting (IBM, 2026). There are three main features that need to be considered when performing a random forest regression: `n_estimators` (number of trees in the random forest model), `max_depth` (the maximum depth of the decision tree), and `min_samples_split` (the minimum number of samples needed to split an internal node) (IBM, 2026) (scikitlearn, 2026). These three parameters were optimized during the hyperparameter tuning process of this model. The metrics used to evaluate the model were RMSE, MAE, and R^2 . RMSE heavily penalizes outliers in the dataset. MAE reports the average error in the dataset. Both RMSE and MAE need to be minimized to reduce error. R^2 measures the coefficient of determination and needs to be maximized. These three metrics will provide a clear picture of the strength of the random forest model.

For the agriculture dataset, the baseline random forest model reported a higher RMSE, MAE and lower R^2 than the hyperparameter tuned models. The random search hyperparameter tuning investigated `n_estimators` (10, 33, 100), `max_depth` (2,4,none), and `min_samples_split` (2,3,4) randomly and outputs the best parameters using negative RMSE as a baseline for

improvement. The random search marginally improved on all three metrics compared to the baseline model. Grid Search was then performed optimizing `n_estimators` (10, 33, 100), `max_depth` (2,4,none), and `min_samples_split` (2,3,4) by testing each possible combination of values and outputs the best parameters using negative RMSE. The grid search marginally underperformed the random search model, but still outperformed the baseline model. The full tabulated results for the agriculture dataset can be found in FIG. 7.

The CO2 emissions dataset was also evaluated using random forest in the same way as the agricultural dataset, employing the same methods for hyperparameter tuning. The Random Search method performed worse than the baseline random forest model. The grid search model outperformed both the baseline model and the random search model. The results for the CO2 emissions dataset can be found in FIG.8.

Overfitting

For most of the analysis, test train split was used to prevent overfitting. This technique sets aside a portion of the data to work as unseen data, and thus disallows your model to memorize the entire dataset and make all of its predictions on the entire dataset alone. After training, the test data is given to the model. The gap between the model's performance on the training set and the test set allows for hyperparameter tuning. This was utilized in my analysis since the datasets I was working with are large datasets that could be split into smaller subsections. This dataset was both large and complex, so that benefitted test-train-split over other overfitting techniques such as cross validation, which performs well on smaller datasets, and regularization, which is less universal than test-train-split.

Metrics and Hyperparameter Tuning

As discussed in the depth section, the metrics that were used for this analysis were RMSE, R^2 , and MAE. RMSE and R^2 were the primary metrics used for the analysis as the data was meticulously cleaned during exploratory data analysis, removing outliers and columns that were multicollinear. RMSE penalized outliers heavily, so this was used due to the cleanliness of the dataset. R^2 was used as it is a universally powerful predictor of correlation with the target variable. Since we were predicting a particular column from each dataset,

Hyperparameter tuning was used in random forest employing random search and grid search. The parameters searched were `n_estimators`, `max_depth`, and `min_samples_split`. More details regarding random forest hyperparameter tuning are present in the depth section. Tuning was also used for SVM, tuning `c`, `gamma`, and `epsilon`. These were chosen as they are the primary metrics used in SVM. Hyperparameter tuning was not employed for other models, as they were less complex and did not require tuning.

Exploratory Data Analysis

Heatmap data informed the selection of target variables. Variables that had high correlation coefficients were selected for analysis. VIF was also employed to remove variables that introduce multicollinearity. Previous EDA in earlier milestone projects led to removal of other variables that did not correlate with target variables. RMSE was chosen as a metric due to the lack of extreme outliers in the dataset.

Expected VS Unexpected Results

I expected the CO2 emissions dataset to have strong correlations, and all of the models showed high R^2 values and low error. Each of the models performed well with the CO2 emissions except the SVM model. This did not work with the SVM dataset, and further analysis

may be required here. I believe that less complex models worked better than the SVM model due to the strong correlation of the variables with the target variable. The agriculture dataset had less strong results. This is what I expected since climate data is significantly more finicky than other data types. Climate is notoriously difficult to predict and is influenced by many factors. The best performing model was random forest, which exemplifies that the relationships between the variables are complex and require complex models to draw meaningful conclusions. I am interested in exploring this dataset with more advanced machine learning algorithms.

Figures

FIG. 1: Polynomial Linear Regression Results
Temperature Change from CO2

Degree	Train RMSE	Test RMSE	Train R²	Test R²
1	0.0161	0.0261	0.9755	0.8656
2	0.0135	0.0294	0.9826	0.8284
3	0.0099	0.0202	0.9906	0.9189

FIG, 1: Polynomial Regression for CO2 Emissions Dataset

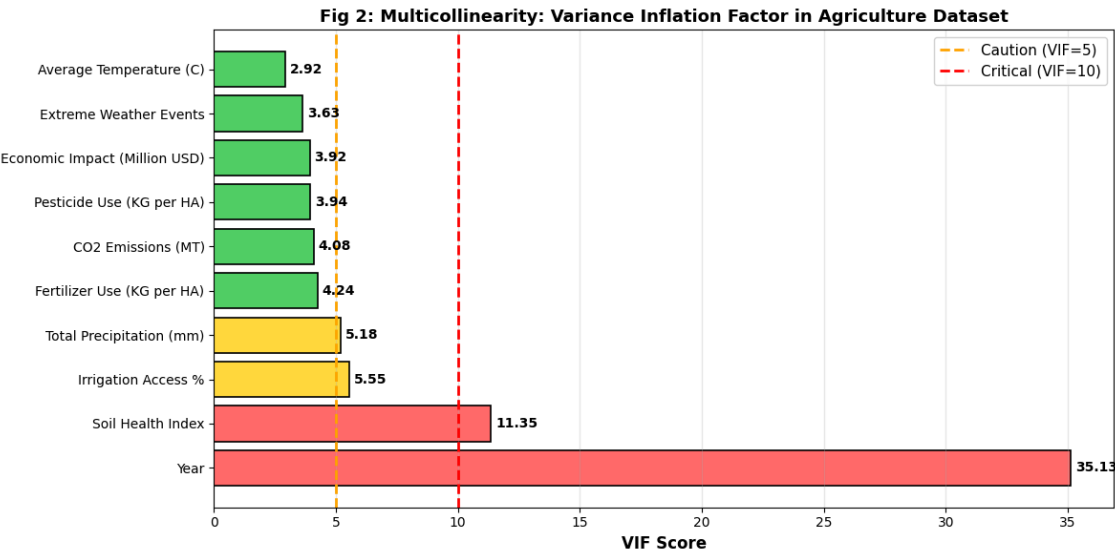


FIG. 2: VIF analysis for Agricultural Dataset.

FIG. 3: Climate Change and Agriculture: Model Comparison

	Lasso	Ridge	Elastic Net
RMSE	0.6643	0.6645	0.6643
R²	0.5274	0.5272	0.5274

FIG. 3: Lasso, Ridge, and Elastic Net regressions on the Agricultural Dataset

FIG. 4: Agriculture PCR vs PLSR Performance Comparison

Method	Train RMSE	Test RMSE	Train R ²	Test R ²
PCR	0.7459	0.727	0.3959	0.4339
PLSR	0.6753	0.6645	0.5048	0.5271

FIG. 4: Agricultural Dataset PCR vs PSLR comparison.

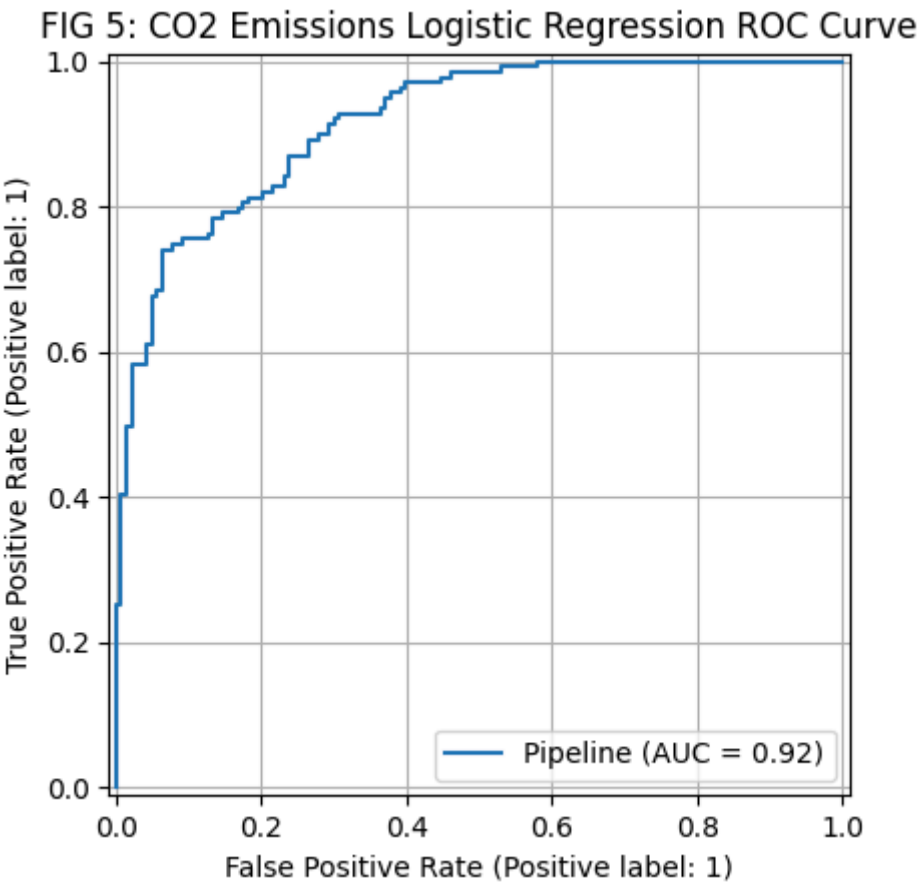


FIG. 5: CO2 Emissions ROC curve for Logistic Regression

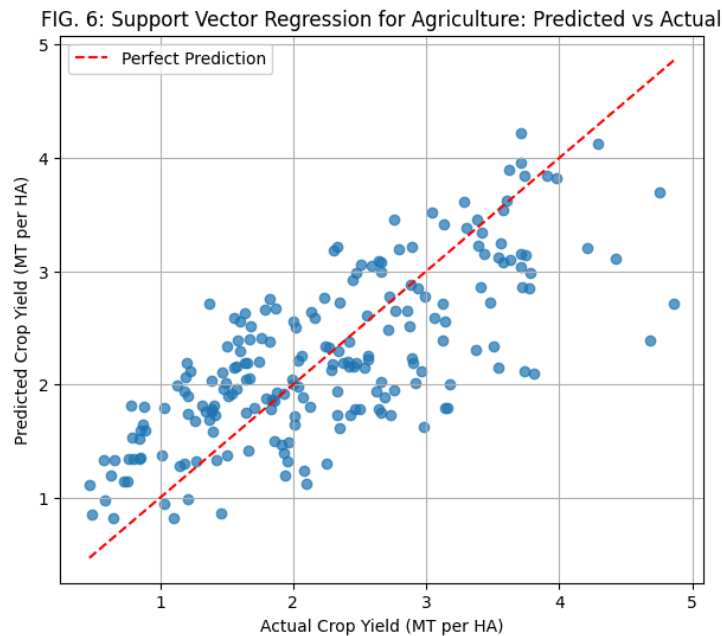


FIG. 6: Support Vector for Agriculture dataset.

FIG. 7: Agriculture vs Climate Change: Random Forest Model Comparison

	Baseline RF Model	Random Search	Grid Search
RMSE	0.6542	0.6403	0.6408
MAE	0.5367	0.5261	0.5275
R ²	0.5416	0.5609	0.5603

FIG. 7: Agriculture dataset Random Forest Model and Optimizations

FIG. 8: CO2 Emissions Per Capita: Random Forest Model Comparison

	Baseline RF Model	Random Search	Grid Search
RMSE	0.0337	0.0364	0.0334
MAE	0.0039	0.0046	0.0037
R ²	0.7756	0.7374	0.7797

FIG. 8: CO2 Emissions Random Forest Model and Optimizations

AI Appendix

AI was used to assist with coding during the writing of this paper. The link to my coding session is here: **<https://terriergpt.bu.edu/share/xcfnH1bqiCjOrahmf7Cja>**

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